

Conventions and Notation

Sandro Vitenti

0.1 Scope

This page fixes, once and for all, the units, metric, and index conventions used throughout the electrodynamics notes. Every other page in this course assumes these conventions; when a formula elsewhere looks unfamiliar, check here first.

1 Units

Unless otherwise stated, SI units are used throughout. The vacuum constants satisfy

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}.$$

2 Metric

The Minkowski metric is

$$\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1),$$

so that

$$x^2 = \eta_{\mu\nu} x^\mu x^\nu = -(x^0)^2 + \mathbf{x}^2.$$

Greek indices run from 0 to 3, while Latin indices run from 1 to 3. Repeated indices are summed.

3 Coordinates

We write

$$x^\mu = (ct, \mathbf{x}), \quad \partial_\mu = \frac{\partial}{\partial x^\mu}.$$

The d'Alembertian is

$$\square = \partial_\mu \partial^\mu.$$

For two spacetime points,

$$\Delta x^\mu = x^\mu - y^\mu,$$

with

$$(\Delta x)^2 = \eta_{\mu\nu} \Delta x^\mu \Delta x^\nu.$$

4 Electromagnetic field

The electromagnetic tensor is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Its components are

$$F^{0i} = \frac{E^i}{c}, \quad F^{ij} = \epsilon^{ijk} B_k.$$

The four-potential is

$$A^\mu = \left(\frac{\phi}{c}, \mathbf{A} \right).$$

The four-current is

$$J^\mu = (c\rho, \mathbf{J}).$$

5 Maxwell equations

The inhomogeneous equations are

$$\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu.$$

The homogeneous equations are

$$\partial_{[\alpha} F_{\beta\gamma]} = 0.$$

In the Lorenz gauge,

$$\partial_\mu A^\mu = 0,$$

the field equations reduce to

$$\square A^\mu = -\mu_0 J^\mu.$$

6 Energy-momentum tensor

$$T^{\mu\nu} = \frac{1}{\mu_0} \left(F^{\mu\alpha} F^\nu{}_\alpha - \frac{1}{4} \eta^{\mu\nu} F^{\alpha\beta} F_{\alpha\beta} \right),$$

with

$$T^{00} = u = \frac{1}{2\mu_0} \left(\frac{E^2}{c^2} + B^2 \right), \quad T^{0i} = \frac{S^i}{c}, \quad \mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}.$$

7 Particle kinematics

The worldline is $y^\mu(\tau)$, and the proper time satisfies

$$c^2 d\tau^2 = -ds^2.$$

The four-velocity is

$$u^\mu = \frac{dy^\mu}{c d\tau} = \gamma \left(1, \frac{\mathbf{v}}{c} \right),$$

with

$$u^\mu u_\mu = -1.$$

The four-acceleration is

$$a^\mu = \frac{du^\mu}{d\tau},$$

and obeys

$$u_\mu a^\mu = 0.$$

A dot is reserved for the derivative with respect to coordinate time t , e.g. $\dot{\mathbf{v}} = d\mathbf{v}/dt$; the proper-time derivative above is always written out or given its own symbol (a^μ), never as $\dot{u}^\mu$.

8 Useful identities

The retarded Green function satisfies

$$\square G_R(x - x') = \delta^{(4)}(x - x').$$

The point-particle current is

$$J^\mu(x) = cq \int u^\mu(\tau) \delta^{(4)}(x - y(\tau)) \, \mathrm{d}\tau.$$

Useful distribution identity:

$$\delta(f(x)) = \sum_i \frac{\delta(x - x_i)}{|f'(x_i)|}.$$

Feel free to create issues, ask questions, or suggest improvements in the [GitHub repository](#).